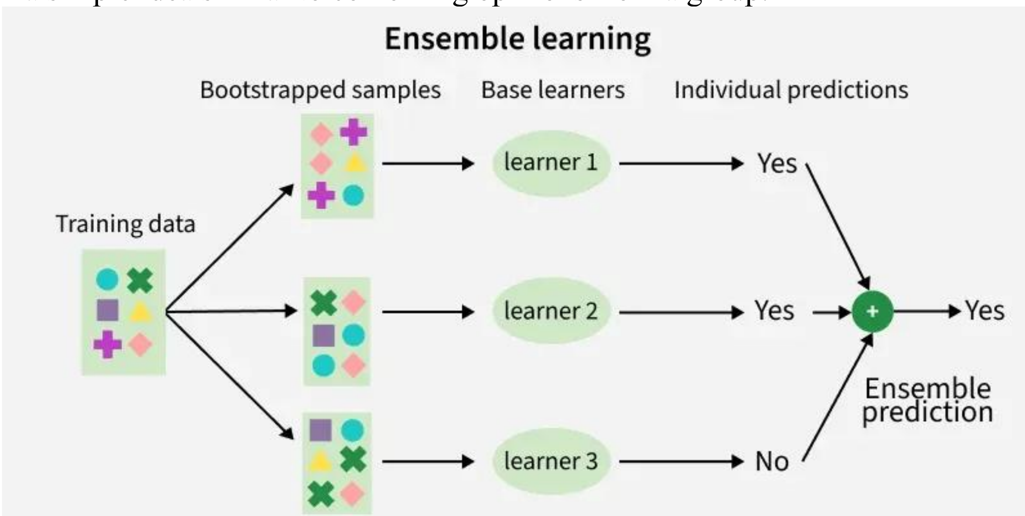
Q1. What is Ensemble Learning technique? What are the types of it?

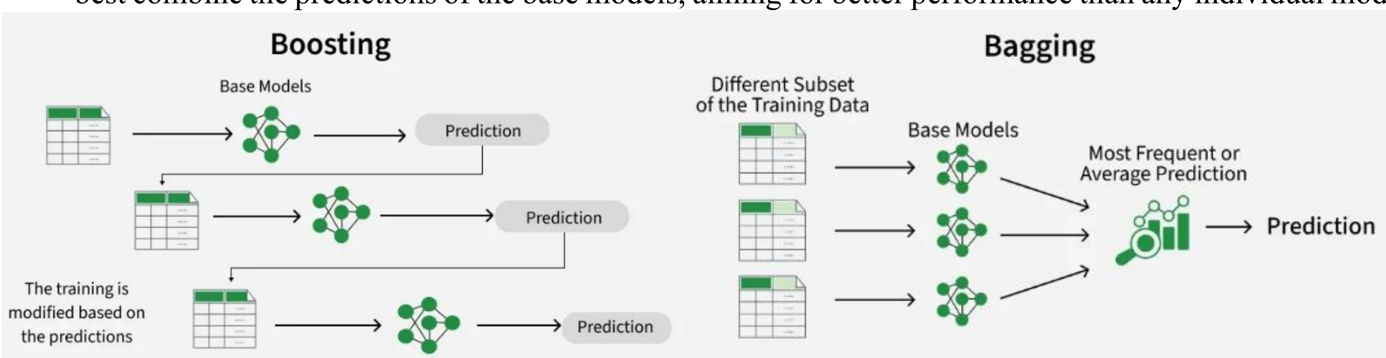
Ensemble learning is a method where multiple models are combined instead of using just one. Even if individual models are weak, combining their results gives more accurate and reliable predictions.

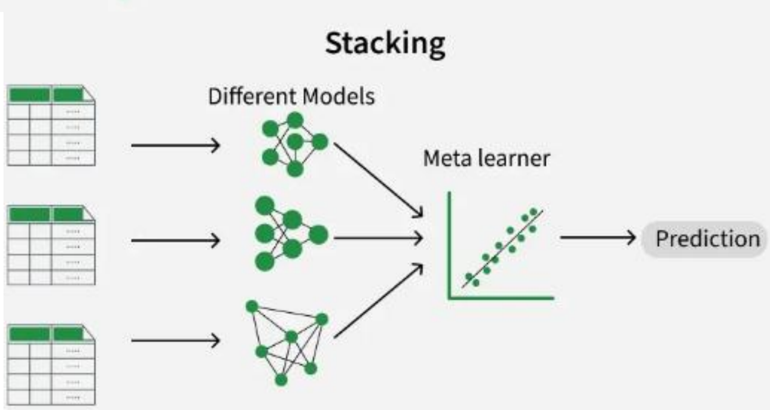
- Uses multiple models together to improve overall accuracy.
- Reduces errors by balancing mistakes across models.
- Works on a simple idea similar to combining opinions from a group.



Types of Ensemble Learning

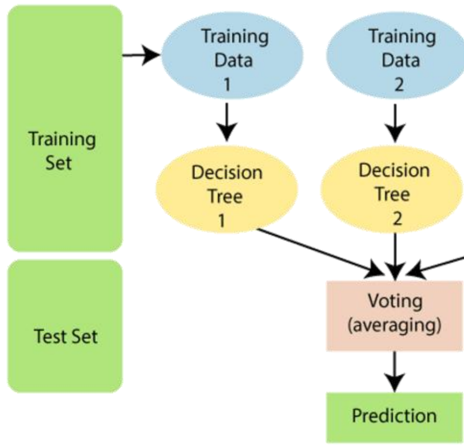
1. **Bagging (Bootstrap Aggregating):** Models are trained independently on different random subsets of the training data. Their results are then combined—usually by averaging (for regression) or voting (for classification). This helps reduce variance and prevents overfitting.
2. **Boosting:** Models are trained one after another. Each new model focuses on fixing the errors made by the previous ones. The final prediction is a weighted combination of all models, which helps reduce bias and improve accuracy.
3. **Stacking (Stacked Generalization):** Multiple different models (often of different types) are trained and their predictions are used as inputs to a final model, called a meta-model. The meta-model learns how to best combine the predictions of the base models, aiming for better performance than any individual model.





Q2. What is Random Forest Algorithm? How does it work? Explain with a real-life example.

Random Forest is a classifier that comprises a number of decision trees on various subsets of the provided dataset and takes the average to enhance the predicted accuracy of that dataset. Instead of depending on a single decision tree, the random forest collects the predictions from each tree and predicts the final output based on the majority vote of predictions.



Why use Random Forest?

Here are some reasons why we should utilize the Random Forest algorithm:

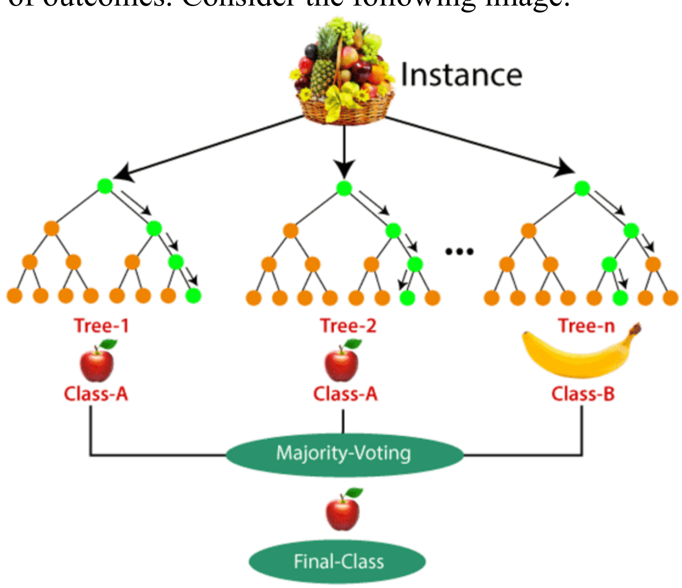
- It requires shorter training time than other algorithms.
- It predicts output with great accuracy, and it works efficiently even on big datasets.
- It can also retain accuracy when a significant amount of data is absent.

How does the Random Forest Algorithm Work?

Random Forest operates in two stages: the first is to generate the random forest by mixing N decision trees, and the second is to make predictions for each tree generated in the first phase.

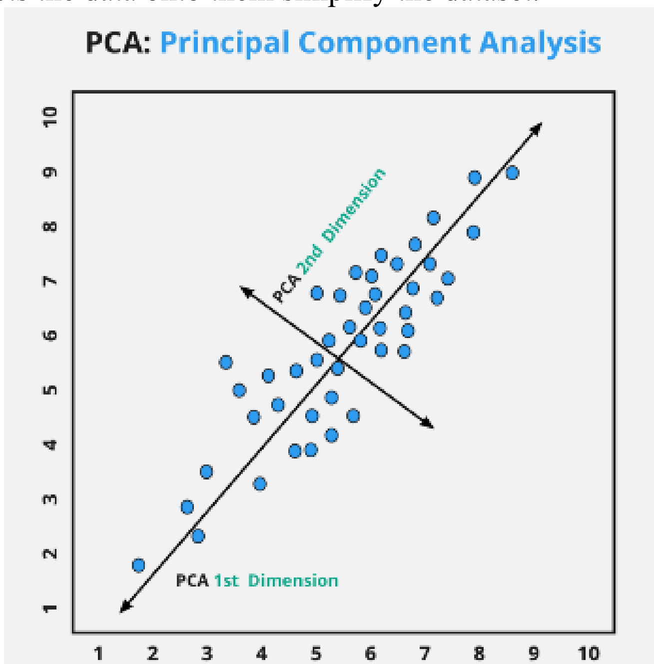
Step 1: Choose K data points at random from the training set.
Step 2: Create decision trees for the specified data points (Subsets).
Step 3: Determine the number N for the number of decision trees you wish to construct.
Step 4: Repeat Steps 1 and 2.
Step 5: Find the predictions of each decision tree for new data points and assign the new data points to the category with the most votes.

The following random forest algorithm example will help you understand how the algorithm works:
Assume there is a dataset containing several fruit photos. As a result, the Random Forest classifier is given this dataset. The dataset is subdivided and distributed to each decision tree. During the training phase, each decision tree gives a prediction result, and when a new data point occurs, the Random Forest classifier predicts the final choice based on the majority of outcomes. Consider the following image:



➢ Principal Component Analysis (PCA)

PCA (Principal Component Analysis) is a dimensionality reduction technique and helps us to reduce the number of features in a dataset while keeping the most important information. It changes complex datasets by transforming correlated features into a smaller set of uncorrelated components. PCA uses linear algebra to transform data into new features called principal components. It finds these by calculating eigenvectors (directions) and eigenvalues (importance) from the covariance matrix. PCA selects the top components with the highest eigenvalues and projects the data onto them simplify the dataset.



PCA converts the original features into new features called **principal components**.
These components are:

- **Uncorrelated**
- **Capture maximum variance**
- **Ordered** (PC1 captures the most variance, PC2 the second most, etc.)

PCA works by finding directions in the data where the data varies the most and projecting the data onto those directions.

🎯 When to Use PCA

1. To reduce dimensionality: Useful when you have many features and want to keep only the most important ones.
2. To avoid the curse of dimensionality: High-dimensional data can cause overfitting; PCA reduces dimensions safely.
3. To remove multicollinearity: If features are highly correlated, PCA combines them into fewer independent components. This helps linear models like logistic regression or linear regression.
4. For visualization: PCA can convert high-dimensional data into 2D or 3D for easy visualization.
5. To speed up machine learning models: Fewer features → faster training.
6. For noise reduction: PCA filters out noise by keeping only the components with high variance.

🔍 What is **Curse of Dimensionality?**

The **Curse of Dimensionality** refers to the problems that occur when the number of features (dimensions) in a dataset becomes very large.
As the number of dimensions increases:

- Data becomes sparse (spread out).
- Finding similar data points becomes difficult.
- Machine learning models require much more data and computation.
- Model accuracy often decreases.

- **Step to solve numerical on PCA:**

Step – 1: Find $\bar{x}$ and $\bar{y}$
Step – 2: calculate covariance matrix (x - $\bar{x}$) (x - $\bar{x}$) using a simple table.
Step – 3: calculate covariance matrix (y - $\bar{y}$) (y - $\bar{y}$)using a simple table.
Step – 4: calculate covariance matrix (x - $\bar{x}$) (y - $\bar{y}$)using a simple table.

Step – 5: covariance matrix

$$cov(x,y) = \sum_{i=1}^{n} \frac{(x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

Step – 6: covariance matrix

$$c = \begin{bmatrix} cov(x,x) & cov(x,y) \\ cov(y,x) & cov(y,y) \end{bmatrix}$$

Step – 7: find eigen values

$$C = \lambda I$$

Step – 8: find eigen vectors

$$CV = \lambda V$$

Step – 9: project the data of principle component on graph.

Q3. Apply PCA on the following data and find the first principal component.

X	2.5	0.5	2.2	1.9	3.1	2.3	2	1	1.5	1.1
Y	2.4	0.7	2.9	2.2	3	2.7	1.6	1.1	1.6	0.9

Solution:

Point	**X**	**Y**
1	2.5	2.4
2	0.5	0.7
3	2.2	2.9
4	1.9	2.2
5	3.1	3.0
6	2.3	2.7
7	2.0	1.6
8	1.0	1.1
9	1.5	1.6
10	1.1	0.9

Number of samples,

$$n = 10$$

Step 1: Find the Mean ($\bar{x}$ and $\bar{y}$)

Mean of X

$$\bar{x} = \frac{2.5 + 0.5 + 2.2 + 1.9 + 3.1 + 2.3 + 2 + 1 + 1.5 + 1.1}{10} = \frac{18.1}{10} = 1.81$$

Mean of Y

$$\bar{y} = \frac{2.4 + 0.7 + 2.9 + 2.2 + 3 + 2.7 + 1.6 + 1.1 + 1.6 + 0.9}{10} = \frac{19.1}{10} = 1.91$$

Therefore,

$$\boxed{\bar{x} = 1.81,\ \bar{y} = 1.91}$$

Step 2: Calculate Covariance Matrix (X, X)

First calculate

$$\left(x - \bar{x}\right)$$

Then square it.

X	X - $\bar{x}$	**(x-$\bar{x}$)^2**
2.5	0.69	0.4761
0.5	-1.31	1.7161
2.2	0.39	0.1521
1.9	0.09	0.0081
3.1	1.29	1.6641
2.3	0.49	0.2401
2.0	0.19	0.0361
1.0	-0.81	0.6561
1.5	-0.31	0.0961
1.1	-0.71	0.5041

Sum

$$\sum (x - \bar{x})^2 = 5.549$$

Now,

$$\boxed{\operatorname{cov}(x,x) = \frac{5.549}{9} = 0.6166}$$

Step 3: Calculate Covariance Matrix (Y, Y)

Y	Y - $\bar{y}$	**($(y-\bar{y})^2$)**
2.4	0.49	0.2401
0.7	-1.21	1.4641
2.9	0.99	0.9801
2.2	0.29	0.0841
3.0	1.09	1.1881
2.7	0.79	0.6241
1.6	-0.31	0.0961
1.1	-0.81	0.6561
1.6	-0.31	0.0961
0.9	-1.01	1.0201

Sum

$$\sum (y-\bar{y})^2 = 6.449$$

Therefore,

$$\boxed{\operatorname{cov}(y,y) = \frac{6.449}{9} = 0.7166}$$

Step 4: Calculate Covariance Matrix (X, Y)
Multiply

$$\left(x-\bar{x}\right)\left(y-\bar{y}\right)$$

Point	**$(x-\bar{x})^2$**	**($(y-\bar{y})^2$)**	**Product**
1	0.69	0.49	0.3381
2	-1.31	-1.21	1.5851
3	0.39	0.99	0.3861
4	0.09	0.29	0.0261
5	1.29	1.09	1.4061
6	0.49	0.79	0.3871
7	0.19	-0.31	-0.0589
8	-0.81	-0.81	0.6561
9	-0.31	-0.31	0.0961
10	-0.71	-1.01	0.7171

Sum

$$\sum (x-\bar{x})(y-\bar{y}) = 5.539$$

Step 5: Find Covariance: Formula

$$\text{cov}(x, y) = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Substitute,

$$= \frac{5.539}{9}$$

$$\boxed{\text{cov}(x, y) = 0.6154}$$

Since covariance is symmetric,

$$\boxed{\text{cov(y, x)} = 0.6154}$$

Step 6: Covariance Matrix

$$C = \begin{bmatrix} cov(x, x) & cov(x, y) \\ cov(y, x) & cov(y, y) \end{bmatrix}$$

Substitute the values,

$$\boxed{C = \begin{bmatrix} 0.6166 & 0.6154 \\ 0.6154 & 0.7166 \end{bmatrix}}$$

Step 7: Find Eigenvalues: The eigenvalues are found by solving

$$| C - \lambda I | = 0$$

Where, C= covariance matrix, I= Identity matrix

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Multiply λ with the identity matrix

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

Subtract λI from C

$$\begin{vmatrix} 0.6166 - \lambda & 0.6154 \\ 0.6154 & 0.7166 - \lambda \end{vmatrix} = 0$$

Expand,

$$(0.6166 - \lambda)(0.7166 - \lambda) - (0.6154)^2 = 0$$

This gives

$$\lambda^2 - 1.3332\lambda + 0.0626 = 0$$

Solve the quadratic equation:

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Here,

$$a = 1,\ b = -1.3332,\ c = 0.0631$$

Substituting,

$$\lambda = \frac{1.3332 \pm \sqrt{(1.3332)^2 - 4(1)(0.0631)}}{2}$$

After calculation,

$$\boxed{\lambda_1 = 1.284}, \boxed{\lambda_2 = 0.049}$$

The larger eigenvalue is

$$\boxed{\lambda_1 = 1.284}$$

This means **the first principal component contains the maximum variance**, so we will use it as the new axis.

Step 8: Find Eigenvector
Solve

$$CV = \lambda V$$

Move everything to one side

$$(C - \lambda I)V = 0$$

Since

$$\lambda = 1.284$$

we have

$$C - \lambda I = \begin{bmatrix} 0.6166 - 1.284 & 0.6154 \\ 0.6154 & 0.7166 - 1.284 \end{bmatrix}$$

Subtract,

$$= \begin{bmatrix} -0.6674 & 0.6154 \\ 0.6154 & -0.5674 \end{bmatrix}$$

Multiply with the unknown vector

$$\begin{bmatrix} -0.6674 & 0.6154 \\ 0.6154 & -0.5674 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives two equations:
Equation (1)

$$-0.6674v_1 + 0.6154v_2 = 0$$

Equation (2)

$$0.6154v_1 - 0.5674v_2 = 0$$

Both equations represent the same line, so solving either one is sufficient.

$$0.6154v_2 = 0.6674v_1$$

Divide both sides by 0.6154:

$$v_2 = \frac{0.6674}{0.6154} v_1$$

$$v_2 = 1.0845v_1$$

Assume

$$v_1 = 1$$

Then

$$v_2 = 1.0845$$

So, the (unnormalized) eigenvector is

$$\begin{bmatrix} 1 \\ 1.0845 \end{bmatrix}$$

Normalize the vector

PCA uses unit vectors, meaning the vector length should be 1.

Compute its length:

$$\sqrt{1^2 + 1.0845^2} = \sqrt{2.176} \approx 1.475$$

Divide each component by 1.475:

$$v_1 = \frac{1}{1.475} = 0.677$$

$$v_2 = \frac{1.0845}{1.475} = 0.736$$

Therefore,

$$\boxed{V = \begin{bmatrix} 0.677 \\ 0.736 \end{bmatrix}}$$

Are the eigen vectors.

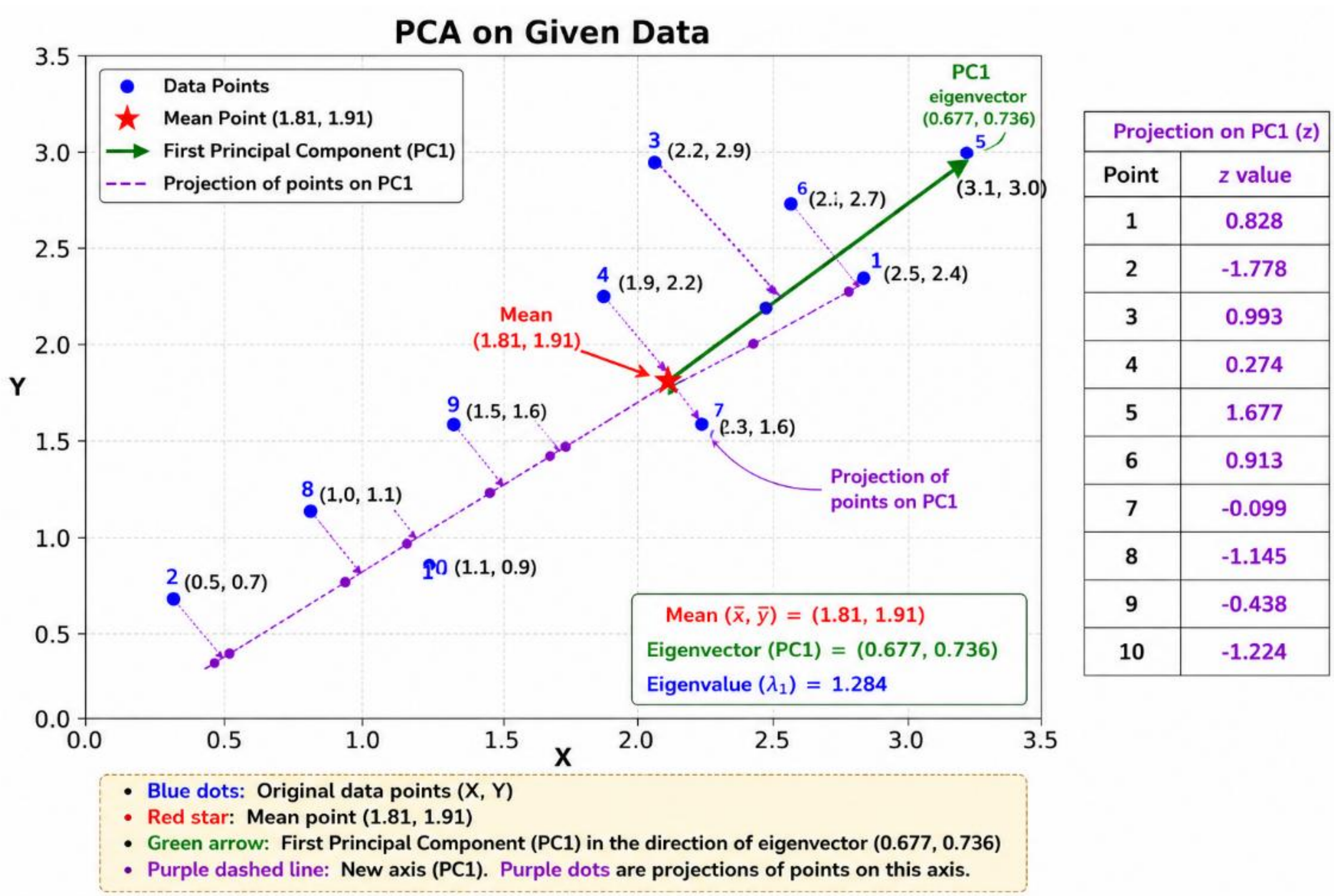
Step 9: Project Data on the Principal Component

The projection formula is

$$z = \left(x - \bar{x}\right) \times 0.677 + \left(y - \bar{y}\right) \times 0.736$$

Let's calculate all the points.

Point	X	Y	$x - \bar{x}$	$y - \bar{y}$	Projection (z)
1	2.5	2.4	0.69	0.49	$(0.69)(0.677) + (0.49)(0.736) = 0.828$
2	0.5	0.7	-1.31	-1.21	$(-1.31)(0.677) + (-1.21)(0.736) = -1.778$
3	2.2	2.9	0.39	0.99	$(0.39)(0.677) + (0.99)(0.736) = 0.993$
4	1.9	2.2	0.09	0.29	$0.09 \times 0.677 + 0.29 \times 0.736 = 0.274$
5	3.1	3.0	1.29	1.09	$1.29 \times 0.677 + 1.09 \times 0.736 = 1.677$
6	2.3	2.7	0.49	0.79	$0.49 \times 0.677 + 0.79 \times 0.736 = 0.913$
7	2.0	1.6	0.19	−0.31	$0.19 \times 0.677 - 0.31 \times 0.736 = -0.099$
8	1.0	1.1	−0.81	−0.81	$-0.81 \times 0.677 - 0.81 \times 0.736 = -1.145$
9	1.5	1.6	−0.31	−0.31	$-0.31 \times 0.677 - 0.31 \times 0.736 = -0.438$
10	1.1	0.9	−0.71	−1.01	$-0.71 \times 0.677 - 1.01 \times 0.736 = -1.224$



Q4. Apply PCA on the following data and find the first principal component.

X	2	3	4	5	6	7
Y	1	5	3	6	7	8

Solution:

Point	**X**	**Y**
1	2	1
2	3	5
3	4	3
4	5	6
5	6	7
6	7	8

Number of samples,

$$n = 6$$

Step 1: Find the Mean $\bar{x}$ and $\bar{y}$

Mean of X

$$\bar{x} = \frac{2+3+4+5+6+7}{6}$$
$$= \frac{27}{6} = 4.5$$

Mean of Y

$$\bar{y} = \frac{1+5+3+6+7+8}{6}$$
$$= \frac{30}{6} = 5$$

Therefore,

$$\boxed{\bar{x} = 4.5,\ \bar{y} = 5}$$

Step 2: Calculate Covariance Matrix (X, X)

Calculate

$$(x - \bar{x})^2$$

X	$(x-\bar{x})$	$(x-\bar{x})^2$
2	-2.5	6.25
3	-1.5	2.25
4	-0.5	0.25
5	0.5	0.25
6	1.5	2.25
7	2.5	6.25

Sum,

$$\sum (x-\bar{x})^2 = 17.5$$

Now,

$$\boxed{\mathrm{cov}(x,x) = \frac{17.5}{6-1} = \frac{17.5}{5} = 3.5}$$

Step 3: Calculate Covariance Matrix (Y, Y)

Y	$(y - \bar{y})$	
1	-4	16
5	0	0
3	-2	4
6	1	1
7	2	4
8	3	9

Sum,

$$16 + 0 + 4 + 1 + 4 + 9 = 34$$

Therefore,

$$\boxed{\operatorname{cov}(y, y) = \frac{34}{5} = 6.8}$$

Step 4: Calculate Covariance Matrix (X, Y)

Find

$$(x - \bar{x})(y - \bar{y})$$

Point	$(x - \bar{x})$	$(y - \bar{y})$	**Product**
1	-2.5	-4	10
2	-1.5	0	0
3	-0.5	-2	1
4	0.5	1	0.5
5	1.5	2	3
6	2.5	3	7.5

Sum,

$$10 + 0 + 1 + 0.5 + 3 + 7.5 = 22$$

Step 5: Find Covariance

Using

$$\operatorname{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

Substitute,

$$= \frac{22}{5} = 4.4$$

Since covariance is symmetric,

$$\boxed{\operatorname{cov}(x, y) = \operatorname{cov}(y, x) = 4.4}$$

Step 6: Construct the Covariance Matrix

$$C = \begin{bmatrix} \text{cov}(x,x) & \text{cov}(x,y) \\ \text{cov}(y,x) & \text{cov}(y,y) \end{bmatrix}$$

Substituting the values,

$$\boxed{C = \begin{bmatrix} 3.5 & 4.4 \\ 4.4 & 6.8 \end{bmatrix}}$$

Step 7: Find the Eigenvalues
We solve

$$|\, C - \lambda I \,| = 0$$

Substitute the covariance matrix,

$$\begin{vmatrix} 3.5-\lambda & 4.4 \\ 4.4 & 6.8-\lambda \end{vmatrix} = 0$$

Using

$$ad - bc$$

we get

$$(3.5-\lambda)(6.8-\lambda) - 4.4^2 = 0$$

Expand,

$$23.8 - 10.3\lambda + \lambda^2 - 19.36 = 0$$

Simplify,

$$\boxed{\lambda^2 - 10.3\lambda + 4.44 = 0}$$

Using the quadratic formula,

$$\lambda = \frac{10.3 \pm \sqrt{10.3^2 - 4(4.44)}}{2}$$
$$= \frac{10.3 \pm 9.40}{2}$$

Hence,

$$\boxed{\lambda_1 \approx 9.85}, \boxed{\lambda_2 \approx 0.45}$$

The larger eigenvalue is

$$\boxed{\lambda_1 = 9.85}$$

Step 8: Find the Eigenvector (First Principal Component)
Use

$$(C - \lambda I)V = 0$$

For

$$\lambda = 9.85$$
$$\begin{bmatrix} 3.5-9.85 & 4.4 \\ 4.4 & 6.8-9.85 \end{bmatrix} = \begin{bmatrix} -6.35 & 4.4 \\ 4.4 & -3.05 \end{bmatrix}$$

Let

$$V = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Then,

$$-6.35v_1 + 4.4v_2 = 0$$

Therefore,

$$v_2 = \frac{6.35}{4.4} v_1$$
$$v_2 = 1.443 v_1$$

Assume

Then

$$v_1 = 1$$

$$v_2 = 1.443$$

The unnormalized eigenvector is

$$\begin{bmatrix} 1 \\ 1.443 \end{bmatrix}$$

Normalize it.
Length,

$$\sqrt{1^2 + 1.443^2} = \sqrt{3.082} = 1.756$$

Divide each value by 1.756,

$$v_1 = \frac{1}{1.756} = 0.569$$

$$v_2 = \frac{1.443}{1.756} = 0.822$$

Therefore,

$$\boxed{V = \begin{bmatrix} 0.569 \\ 0.822 \end{bmatrix}}$$

This is the first principal component (PC1).

Step 9: Project the Data onto PC1
The projection formula is

$$z = \left(x - \bar{x}\right)(0.569) + \left(y - \bar{y}\right)(0.822)$$

Point	$\left(x - \bar{x}\right)$	$\left(y - \bar{y}\right)$	**Projection (z)**
1	-2.5	-4	−4.715
2	-1.5	0	−0.854
3	-0.5	-2	−1.929
4	0.5	1	1.107
5	1.5	2	2.499
6	2.5	3	3.891